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Plastic bioremediation potential of groundwater microbiomes

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Abstract

Global increase in the production of highly recalcitrant plastic materials and their accumulation in the environment are creating the urgent need to develop new sustainable strategies to mitigate this type of pollution. Microbial biotechnology, including blue biotechnology, is a promising option for the sustainable disposal of plastic waste. Despite both soil and water representing potential sources of novel biocatalysts, studies unraveling the biotechnological potential of soil microbial communities considerably outnumber studies of aquatic microbiota. Although traditionally focused on marine resources, the scope of blue biotechnology has expanded to include all aquatic ecosystems and the focus is shifting towards underexplored aquatic environments. Environmental niches that are not readily accessible for sampling, representing reservoirs of enzymatic activities with potential biotechnological applications, such as subsurface environments, especially its core component groundwater, are still relatively understudied. Despite a proposed overlap between microbial enzymatic capacity to degrade one of the substrates that has been available on Earth for a very long time—lignocellulosic biomass and a relatively newly available substrate, introduced only about a century ago by humans—plastics, studies investigating potential for the transformation of fossil and bio-based plastics by lignocellulose-degrading microorganisms and their consortia and bioprospecting for microorganisms capable of degrading both, are still relatively scarce. Here, we discuss and summarize recent advances in decoding plastolytic and lignocellulolytic potential of relatively rarely examined groundwater bacteria.

Keywords Alkaliphiles, Thermophiles, Brackish-saline groundwater microbiomes, Plastolytic Potential, Lignocellulolytic potential, Blue biotechnology

Introduction

Global increase in the production of highly recalcitrant plastic materials and their accumulation in the environment, with over 400 million tons of plastics produced [1] and approximately 220 million tons of plastic waste generated [2] in 2024 worldwide, are creating the urgent need to develop new sustainable strategies to mitigate this type of pollution. Most of the currently used conventional methods for the plastic waste disposal are inefficient from both the environmental and economic perspective [3]. Global research efforts to develop biotechnological solutions for plastic waste management, based on the molecules produced by the environmental microorganisms and their consortia, particularly enzymes able to degrade complex polymeric substrates structurally similar to plastics, such as hydrolases and oxidoreductases [4], are underway. Microbial biotechnology, including blue biotechnology, has been proposed as a promising option for the sustainable disposal of plastic waste [5]. Multipronged approach, combining low- and high-throughput strategies [6], including bioprospecting, metagenomics and other omics-based approaches, directed evolution and metabolic engineering of

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degrading microbes, represents a step forward for identification of novel plastic-degrading microbes and microbial enzymes and other bioactive molecules involved in biodegradation and achieving circular economy for plastics. Despite a significant progress, microbial biotechnology for plastic waste recycling is, at present, neither broadly integrated into mainstream waste management systems [7, 8] nor widely accepted by plastic producers, users and policymakers [9].

Many plastic degrading microorganisms, acting on a range of plastic substrates and isolated from various environments, have been identified so far. Environmental sites most frequently screened for plastic-degrading microbes include environments exposed to moderate and high levels of prolonged plastic pollution exerting selective pressure on microbial communities. These include landfills, soils, wastewater treatment plants, composting sites, marine habitats [10, 11], hydrocarbon-contaminated environments [12], as well as, more recently, surface freshwater ecosystems, such as lakes and rivers [13, 14]. Despite both soil and water representing promising sources of novel biocatalysts, metagenomic studies unraveling the biotechnological potential of soil microbial communities considerably outnumbered studies of aquatic microbiota in the 2010–2021 period, with aquatic metagenomic studies mainly focusing on marine environments [15]. Studies focusing on environmental niches that are not readily accessible for sampling, representing relatively underexplored reservoirs of enzymatic activities with potential biotechnological applications, such as subsurface environments, especially its core component groundwater, are still relatively scarce and are mainly limited to geothermal springs. The scope of blue biotechnology has expanded to include all aquatic ecosystems and the focus is shifting towards underexplored aquatic environments.

There is a mounting evidence in scientific literature [16–21] that ligninolytic microorganisms, capable of biodegradation of biomass component lignin, a very complex polymer recalcitrant to degradation, could also be effective in degrading certain plastic polymers via ligninolytic enzyme systems [22, 23]. Despite some structural similarities shared between biomass-derived and plastic polymers, the potential for the transformation of fossil and bio-based plastics by lignocellulose-degrading microorganisms and their consortia has not been extensively investigated [24]. Bioprospecting for microorganisms capable of degrading both, that could offer promising solutions for addressing green waste and plastic pollution simultaneously, is still very limited.

Here, we discuss and summarize recent advances in decoding plastolytic and lignocellulolytic potential of relatively rarely examined groundwater microbiota, mainly focusing on bacteria.

Groundwater microorganisms and their biotechnological potential

The terrestrial subsurface, extending from centimeters to kilometers below ground, is composed of rocks, gases, sediments and pores through which groundwater flows. It represents a vast, dark, oligotrophic ecosystem characterized by variable spatial physicochemical conditions and limited energy sources [25], hosting up to 40% of Earth's total microbial biomass [26]. Groundwater physicochemical conditions vary significantly across space and depend on the local geological conditions [25]. Rock weathering and mineral dissolution are the primary natural drivers of groundwater chemistry, with water residence times and anthropogenic activities, such as industrial emissions, agricultural runoffs and waste disposal, also affecting its chemical properties and pH. Groundwater classification based on its physicochemical conditions is arbitrary and context-dependent (e.g., regulatory, industrial or environmental context). Classification criteria used in this work is presented in Supplementary Table S1. Groundwater environments represent an underexplored source of resilient microbial communities, dominated by bacteria and archaea, that are driving vital biogeochemical processes and facilitating the natural breakdown of contaminants in groundwater systems [25]. Some groundwater microorganisms are adapted to extreme environmental conditions (e.g. extreme pH values, elevated temperature, pressure and/or salinity). They are poorly understood, due to the technical difficulties with sampling subterranean habitats and due to the low abundance and physiological tolerance of most subterranean species, often limiting sample size and making laboratory experiments challenging [27].

Groundwater microorganisms have diverse metabolisms that can facilitate bioremediation and produce biomolecules that are active in extreme environments, such as extremozymes, extremolytes, biosurfactants, exopolysaccharides and pigments. Microorganisms and their biomolecules have potential for biotechnological exploitation [28] in various fields such as biotechnology, biodegradation, biofuel production, bioremediation and astrobiology, as well as in food and pharmaceuticals production and agricultural, cosmetics and textile industries [29]. From a biotechnological perspective, thermal and alkaline groundwater environments have been the focus in the seearch for plastic-degrading microorganisms. Microbial plastic degradation in acidic groundwater is not well studied and low pH, although promoting spontaneous hydrolysis of certain polymers, generally hinders

biodegradation processes and slows down both microbial growth and enzymatic activity.

Overview of the selected groundwater microorganisms with recently discovered plastolytic and lignocellulolytic potential is presented in the Fig. 1 and in the Supplementary Table S1.

Metagenomic insights into groundwater microbial plastolytic and lignocellulolytic potential

Omics approaches, including metagenomics, transcriptomics, proteomics and metabolomics, have been employed for characterization of microbial metabolic pathways and adaptations enabling tolerance and biodegradation of different polutants in a range of environmental niches [30]. They have also been used for the identification of genes and enzymes involved in polymer degradation, providing novel microbial resources with potential biotechnological applications [31]. Despite the importance of omics-based approaches for identification of new biotechnologically relevant microorganisms and biomolecules, metagenomic insights into groundwater plastic and lignocellulose degrading potential are still relatively scarce.

Metagenomic study of putative plastic-degrading prokaryotes across 35 New Zealand geothermal springs (pH 4–10, 20–80 °C) confirmed wide distribution of putative plastic-degrading microorganisms and enzymes in geothermal environments [32]. Aprroximately two thirds of investigated springs harboured previously reported putative plastic-degrading bacteria, predominantly belonging to genera *Flavobacterium*, *Pseudomonas*, *Sphingomonas*, and *Rheinheimera*. Genera *Pseudomonas* and *Flavobacterium* dominated all surveyed acidic (30–50 °C) and basic (20–30 °C) springs, respectively, with other taxa showing narrower pH or temperature preferences. The highest diversity of putative plastic degrading bacteria was detected in springs with moderate thermophilic conditions (30–40 °C) and near-neutral pH. Alkaline springs with moderate temperatures were identified as hotspots with the highest abundance and diversity of genes encoding putative plastic-degrading enzymes, including polyurethanases, PHA/PHB depolymerases, alkane monooxygenases, PVA/PEG dehydrogenases, laccases, esterases, lipases and proteases [32].

Study investigating microbial diversity and biomass-degrading potential of 71 publicly available metagenomic datasets from 58 globally distributed hot springs [33] identified a wide diversity of microbes unique to varying temperature and pH ranges. It also catalogued the CAZymes (carbohydrate-active enzymes)-encoding potential, including 19 glycosyl hydrolases putatively involved in cellulose, hemicellulose and oligosaccharide degradation. Study did not identify distinct geochemical conditions, such as specific temperature or pH ranges, as specially favorable for biomass degradation.

Plastolytic and lignocellulolytic potential of brackish-saline groundwater microbiomes

Reserves of continental brackish groundwater account for about 11% of the total water stored within aquifers [34]. Groundwater salinity can originate from both natural and anthropogenic sources [35]. Brackish-saline groundwater microbiomes host unique halophilic and halotolerant microorganisms, with lower microbial diversity compared to freshwater and with variations of microbial communities' composition along a groundwater salinity gradient [36]. Brackish-saline groundwater microbiomes are an important but underexplored potential source of biomolecules with applications in a range of biotechnological and environmental processes [37].

Bioprospecting of microbial plastolytic and lignocellulolytic potential of brackish cold and thermal mildly alkaline (pH 7.9) and slightly acidic (pH 6.5–6.8) groundwaters in Serbia, using a range of plastic and lignocellulosic polymers as sole carbon (C) source for microbial growth, detected isolates with polyurethane (PU)-, polyethylene terephthalate (PET)—and cellulose-degrading potential [38]. Metagenomic 16S rRNA amplicon sequencing of cultivated groundwater samples identified *Aeromonas* and *Exiguobacterium* as the most abundant genera. Selected isolates with plastolytic potential were identified as *Stenotrophomonas*, *Flavobacterium*, *Pantoea*, *Enterobacter*, *Pseudomonas*, *Serratia*, *Acinetobacter*, and *Proteus*, while cellulolytic bacteria belonged to genera *Rhizobium* and *Shewanella*.

Metagenomic analyses of a groundwater sample collected from the deep (875 m), moderately alkaline (pH 8.75) brine aquifer with high salinity (94–315 g/L) in Canada, revealed Proteobacteria as dominant phyla. It also reconstructed 7 MAGs affiliated with *Sphingomonas*, *Acinetobacter*, *Pseudomonas*, and *Massilia* [39]. All MAGs contained a number of putative CAZymes and polyphenol oxidases, capable of degradation of wide range of biomass components, including lignin.

Putative polyurethanolytic bacteria were identified in the two cold brackish mildly alkaline (pH 7.6–7.9) groundwaters in Serbia [40]. The most abundant cultivable bacteria were *Pseudomonas* and *Acidovorax*, while selected isolates with PU-degrading potential were identified as *Stenotrophomonas*, *Pseudomonas*, *Aeromonas*, *Vibrio* and *Acidovorax*. All polyurethanolytic isolates could also grow on all tested lignocellulosic substrates (carboxymethylcellulose–CMC, xylan and lignin) as sole C source without clearance zone formation, indicating incomplete degradation of tested substrates. Another brackish cold mildly alkaline (pH 7.4) spring, dominated

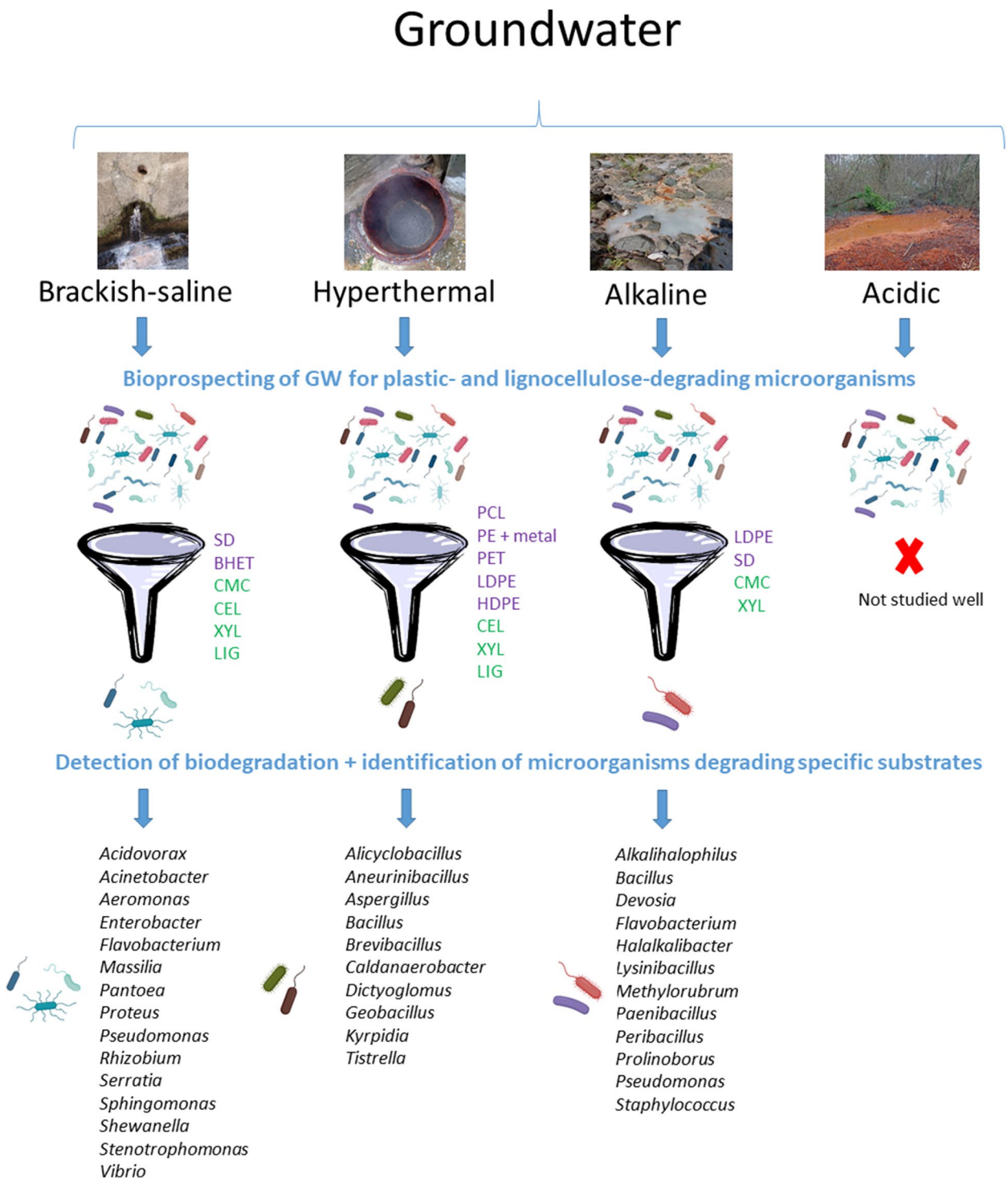

Fig. 1 Bioprospecting of groundwater environments to identify microorganisms with plastolytic and lignocellulolytic potential. Groundwater environments: Brackish-saline (Total Dissolved Solids content > 1 g/L); Hyperthermal (T > 40°C); Alkaline (pH ≥ 7.2); Acidic (pH ≤ 6.8). Plastic (purple) and lignocellulosic (green) degradation substrates: SD—Impranil® DLN-SD; BHET—bis(2-hydroxyethyl) terephthalate; CMC—carboxymethylcellulose; CEL—cellulose; XYL—xylan; LIG—lignin; PCL—poly-ε-caprolactone; PE – polyethylene; PET—polyethylene terephthalate; LDPE—low-density polyethylene; HDPE—high-density polyethylene. Graphical representation of microorganisms is created using https://BioRender.com. Identified plastic- and lignocellulose-degrading microorganisms are presented at genus level

by members of *Aeromonas* genus upon cultivation, harbored cellulolytic bacteria that could also utilize SD and lignin as sole C source [41].

Plastolytic and lignocellulolytic potential of hot spring microbiomes

Hot springs are chemically diverse environments found in a variety of geological settings and distributed worldwide. They act as natural geothermal water discharge points and are associated with zones of higher tectonomagmatic or volcanic activity [42]. Hot springs host an array of thermophilic microorganisms representing a valuable source of thermostable enzymes for biotechnological applications, including a wide range of lignocellulolytic enzymes, most notably hemicellulases [43]. Bacteria isolated from hot springs, including *Bacillus*, *Geobacillus*, *Anoxybacillus*, *Lysinibacillus*, *Paenibacillus*, and *Brevibacillus*, are among thermophiles that are already used as workhorses in biotechnology [44]. However, research on plastic-degrading microbial potential from these environments is lagging.

Comparison of diversity of cultivable bacteria from the plastic-contaminated hot spring in Bulgaria, with and without enrichment on poly-ε-caprolactone (PCL) [45], revealed that *Meiothermus* and *Brevibacillus* are enriched with PCL. Polymer weight loss after 4 weeks was 100% for PCL-enriched natural consortia, 74.9% for *Brevibacillus thermoruber* strain 7 and *Aneurinibacillus thermoaerophilus* strain 2 co-culture and 63.6% for strain 7 alone, with similar PCL degradation intermediate profiles observed for the co-culture and the single strain. Degradation was confirmed by scanning electron microscopy (SEM), with biofilm formation only in co-culture. Thermophilic lipase from *B. thermoruber* strain 7 was later identified as enzyme involved in this relatively quick PCL degradation to monomers [46]. Study analyzing fungal diversity of water and sediment samples collected from the hot spring (42°C) in India [47] identified *Aspergillus versicolor* GSF-A1 as effective in biodegradation of oxo-biodegradable plastic film, composed of polyethylene (PE) and metal salts, achieving 73% plastics degradation in 10 and complete degradation in 20 days. Surface and chemical changes in the plastic film were confirmed by SEM and Fourier Transform Infrared Spectroscopy (FT-IR) analyses. Investigation of PET-degrading bacteria from plastic litter-adhered biofilms in two hot springs (62–66 °C; pH 6.2–6.4) in Peru [44] identified mainly *Bacillus* spp. Two isolates were affiliated to *Brevibacillus borstelensis* and one, surprisingly, to *Tistrella mobilis*, a mesophilic bacterium previously associated with polyhydroxyalkanoate production and hydrocarbon biodegradation. Atanasova and colleagues [48] enriched microorganisms capable of degrading low-density polyethylene (LDPE) and high-density polyethylene (HDPE) from water and sediment samples collected from two hot springs (78–80 °C), proving a first evidence of anaerobic PE degradation by thermophilic bacteria. Biodegradation was confirmed by SEM and transmission electron microscopy (TEM). Comparison of microbial diversity between PE-supplemented cultures after 45 days of incubation and initial hot spring sample revealed enrichment of *Dictyoglomus* and *Caldanaerobacter* on PE, which was supported by the light microscopy of enrichment cultures. Study of water and sediment slurry samples collected from two alkaline geothermal springs in the USA led to the isolation of closely related lignocellulolytic bacteria identified as *Geobacillus thermoleovorans*, most of which could form biofilms on lignin, a phenotype correlated with increased bioconversion [49]. Another study [50] integrated stable isotope probing and genome-resolved metagenomics to investigate the putative degradation mechanisms of the ^{13}C-labeled synthetic lignin by microbial communities from moderately thermophilic (52 °C) Canadian hot spring sediments. Fourteen MAGs significantly enriched with ^{13}C from synthetic lignin belonged to Actinomycetes, Firmicutes and Gammaproteobacteria phyla. They included *Kyrpidia* sp. encoding catechol *meta*-cleavage and *Alicyclobacillus* spp. encoding a suite of putative laccase-like multi-copper oxidases, representing potential enzymatic candidates for the lignin valorization at elevated temperatures.

Alkaline groundwater and alkaliphiles

Alkaline groundwaters are generated primarily through the weathering and hydrolysis of carbonate rock minerals by the deep-circulating meteoric water. They form unique microbiological niche characterized by low microbial diversity and dominance of anaerobes due to the limited availability of oxidants and dissolved inorganic carbon [51]. Globally rare, ophiolite-hosted hyperalkaline waters (pH > 10) are formed in the serpentinization process, which generates highly reducing conditions. Meta-analysis of 70 publicly available metagenomes from nine globally distributed serpentinizing systems showed similar patterns across the entire dataset, regardless of geographical location, suggesting that similar environmental factors, specifically serpentinization processes, shape the microbial community structures and that microbial diversity is inversely correlated with pH [51]. Alkaliphilic and alkalitolerant microorganisms, especially from hyperalkaline groundwaters environments, have a significant biotechnological potential due to their ability to thrive in high-pH environments [52]. Most plastic-degrading bacteria isolated from alkaline environments reported in literature belong to obligate alkaliphiles

within the Bacilliaceae family, with similar pattern also observed for thermophilic plastic-degrading bacteria [53]. Investigation of LDPE film biodegradation by bacteria from serpentinization-driven hyperalkaline spring (pH 11) in Philippines identified 3 LDPE-degrading isolates from genera *Halalkalibacter*, *Alkalihalophilus* and *Bacillus* (previously all classified as *Bacillus*) and one *Prolinoborus* isolate, demonstrating 6.3–9.9% reduction of the LDPE film dry weight after 90 days of incubation, which was confirmed by FT-IR and SEM analyses [54]. Increased effectiveness of the LDPE degradation by a bacterial consortium composed of *A. pseudofirmus* and *Salipaludibacillus agaradhaerens*, enriched from the same alkaline spring, was observed in the presence of the iron oxide nanoparticles [55], with 18.3% and 13.7% polymer weight reduction after 60 days of incubation with consortium in the presence and absence of nanoparticles, respectively. Based on FT-IR and SEM findings, it was hypothesized that nanoparticles enhance biodegradation by enhancing consortium hydrophobicity and increasing its' adhesion to the plastic surface. Bioprospecting of groundwater and sediments from four serpentinization-driven cold hyperalkaline (pH 11.1–11.7) springs in Serbia on a range of plastic and lignocellulosic substrates identified cellulolytic bacteria belonging to *Bacillus*, *Peribacillus*, *Paenibacillus* and *Lysinibacillus* genera [56]. All identified isolates also demonstrated the ability to use PU-model substrates SD and Impranil® DL 2077 (DL) as sole C source for their growth. One of the *Bacillus* isolates could also degrade more complex xylan substrate [57]. Interestingly, 16S rRNA gene sequences of two SD-degrading *Flavobacterium* isolates, one isolated from a nearby hyperalkaline (pH 11.7) groundwater occurrence [58] and another from a brackish mildly acidic groundwater [38], clustered closely together, suggesting a wide range of conditions in which this genus persists in the environment. Investigation of polyurethanolytic activity of microorganisms from the cold mildly alkaline (pH 8.1) abandoned mine drainage, originating from the terminated lead–zinc mine, identified *Pseudomonas*, *Staphylococcus*, *Bacillus*, *Devosia*, and *Methylorubrum* as potential PU-degraders [59].

Future prospects and conclusions

To date, bacteria with biotechnological potential for the degradation of various complex and recalcitrant polymers, including plastics and lignocellulose, have been recorded in diverse groundwater types worldwide. Thermal and alkaline groundwaters are emerging as promising niches in the search for microbial enzymes and metabolites involved in polymer biodegradation processess.

While significant progress has been made in identifying plastic-degrading microbes and, to lesser extent, microbial enzymes and other metabolites involved in biodegradation, blue microbial biotechnology for plastic waste recycling is still in the early stages of implementation. Main bottlenecks for speeding this process up are (i) effectiveness of metagenomic exploration of the environmental microorganisms, representing an excellent method for surveying large datasets and identification of novel microbial candidates for biodegradation of complex polymers, is hindered by underpopulated sequence databases, containing only limited number of taxa and putative proteins deposited; (ii) available microbial catalogues are heavily skewed towards most frequently screened environments and subterenean aquatic environments remain underrepresented in these databases; (iii) a very limited number of functions is characterized in detail. Experimental work including screening for novel polymer-degrading candidates and in-depth validation of biodegradation, feeding public repositories, remains essential for advancing biotechnology and synthetic biology. While data on the distribution of environmental plastic waste worldwide are being increasingly collected, better correlation studies, connecting data on plastic distribution and type with data on microorganisms inhabiting plastisphere and data validating biodegradation of certain types of polymers are much needed. Lastly, expanding screening efforts to novel environments and adopting broad and inclusive bioprospecting strategies will also aid future discovery efforts.

Abbreviations

BHET	Bis(2-hydroxyethyl) terephthalate
CAZymes	Carbohydrate-active enzymes
CMC	Carboxymethylcellulose
DL	Impranil® DL 2077
FT-IR	Fourier Transform Infrared spectroscopy
HDPE	High-density polyethylene
LDPE	Low-density polyethylene
MAG	Metagenome-Assembled Genome
PCL	Poly-ε-caprolactone
PE	Polyethylene
PET	Polyethylene terephthalate
PU	Polyurethane
SD	Impranil® DLN-SD
SEM	Scanning Electron Microscopy
TEM	Transmission Electron Microscopy

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s44314-025-00034-5.

Supplementary Material 1. Supplementary Table S1. Overview of microorganisms with plastolytic and lignocellulolytic potential isolated from various groundwater environments.

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Authors' contributions
MC: conceptualization, investigation, writing—original draft; writing—review and editing. VŠ: conceptualization, investigation, writing—original draft; writing—review and editing. All authors read and approved the final manuscript.

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Data availability
All data generated or analyzed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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